HILB^D OF A CLOSED SUBSCHEME IS A CLOSED SUBSCHEME OF HILB^D

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 340-347 of sources/stacks-pic/pinned-original-source.tex; proof span is lines 349-378, the complete author proof (28 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 10 of queue_pic.json (pb=1637, stmt 340-347, proof 349-378).

Editorial note. stacks-pic wave-1 (pi01): fresh adjudication accept (2026-10-14); pool per standing queue hypercovering -> pic (SMC reassigned to Zhen per Chief 08:5x); pic.tex 27 labels/23 with proof judged 21 accept / 2 reject / 4 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 001L: sources/stacks-pic/pinned-original-source.tex, source lines 378–378. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07AI: sources/stacks-pic/pinned-original-source.tex, source lines 378–378. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $i : X \to Y$ be a closed immersion of schemes. If $\mathrm{Hilb}^d_{Y/S}$ is representable by a scheme, so is $\mathrm{Hilb}^d_{X/S}$ and the corresponding morphism of schemes $\underline{\mathrm{Hilb}}^d_{X/S} \to \underline{\mathrm{Hilb}}^d_{Y/S}$ is a closed immersion.*

Proof. Let T be a scheme over S and let $Z \in \mathrm{Hilb}^d_{Y/S}(T)$. Claim: there is a closed subscheme $T_X \subset T$ such that a morphism of schemes $T' \to T$ factors through T_X if and only if $Z_{T'} \to Y_{T'}$ factors through $X_{T'}$. Applying this to a scheme T_{univ} representing $\mathrm{Hilb}^d_{Y/S}$ and the universal object[1] $Z_{univ} \in \mathrm{Hilb}^d_{Y/S}(T_{univ})$ we get a closed subscheme $T_{univ,X} \subset T_{univ}$ such that $Z_{univ,X} = Z_{univ} \times_{T_{univ}} T_{univ,X}$ is a closed subscheme of $X \times_S T_{univ,X}$ and hence defines an element of $\mathrm{Hilb}^d_{X/S}(T_{univ,X})$.

Date: stacks-project:pic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

[1] See Categories, Section 001L

A formal argument then shows that $T_{univ,X}$ is a scheme representing $\mathrm{Hilb}^d_{X/S}$ with universal object $Z_{univ,X}$.

Proof of the claim. Consider $Z' = X_T \times_{Y_T} Z$. Given $T' \to T$ we see that $Z_{T'} \to Y_{T'}$ factors through $X_{T'}$ if and only if $Z'_{T'} \to Z_{T'}$ is an isomorphism. Thus the claim follows from the very general More on Flatness, Lemma 07AI. However, in this special case one can prove the statement directly as follows: first reduce to the case $T = \mathrm{Spec}(A)$ and $Z = \mathrm{Spec}(B)$. After shrinking T further we may assume there is an isomorphism $\varphi : B \to A^{\oplus d}$ as A-modules. Then $Z' = \mathrm{Spec}(B/J)$ for some ideal $J \subset B$. Let $g_\beta \in J$ be a collection of generators and write $\varphi(g_\beta) = (g^1_\beta, \ldots, g^d_\beta)$. Then it is clear that T_X is given by $\mathrm{Spec}(A/(g^j_\beta))$. $\square$